Modeling Human Reinforcement Learning with Disentangled Visual Representations

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Abstract

Humans are able to learn about the visual world with a remarkable degree of generality and robustness, in part due to attention mechanisms which focus limited resources onto relevant features. Deep learning models that seek to replicate this feature of human learning can do so by optimizing a so-called "disentanglement objective", which encourages representations that factorize stimuli into separable feature dimensions [4]. This objective is achieved by methods such as the β-Variational Autoencoder (β-VAE), which has demonstrated a strong correspondence to neural activity in biological visual representation formation [5]. However, in the β-VAE method, learned visual representations are not influenced by the utility of information, but are solely learned in an unsupervised fashion. In contrast to this, humans exhibit generalization of learning through *acquired equivalence* of visual stimuli associated with similar outcomes [7]. The question of how humans combine utility-based and unsupervised learning in the formation of visual representations is therefore unanswered. The current paper seeks to address this question by developing a modified β-VAE model which integrates both unsupervised learning and reinforcement learning. This model is trained to produce both psychological representations of visual information as well as predictions of utility based on these representations. The result is a model that predicts the impact of changing utility on visual representations. Our model demonstrates a high degree of predictive accuracy of human visual learning in a contextual multi-armed bandit learning task [8]. Importantly, our model takes as input the same complex visual information presented to participants, instead of relying on hand-crafted features. These results provide further support for disentanglement as a plausible learning objective for visual representation formation by demonstrating their usefulness in learning tasks that rely on attention mechanisms.

Keywords: Representation Modelling, Visual Learning, Attention

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1 Introduction

In the context of visual learning, a representation refers to the internal psychological state of an agent while perceiving the external information relevant to a task. In complex learning tasks, agents must form efficient representations of information due to limited cognitive resources by ignoring or abstracting away irrelevancies. An important phenomenon in studies of human representation learning is *acquired equivalence*, which describes increased generalization of dissimilar stimuli based on associations of similar outcomes [7]. The focus of this work is in modelling the visual representations formed by humans in utility-based learning tasks.

One method for modelling visual representation formation in humans is the Variational Auto-Encoder (VAE), a deep neural-network model that forms informationally limited representations of visual information. This is done by taking an input stimuli and reducing the dimensionality of the neural network layer by layer to form a constrained representation, which is then used to reconstruct the original stimuli. In VAEs the learned latent representations take the form of a multi-variate Gaussian distribution. This distribution can be sampled from to form lossy reconstructions of the original stimuli. VAE models are trained to produce reconstructions that are as close to the original stimuli as possible, and thus effectively learn useful low-dimensional representations.

Reinforcement Learning (RL) techniques for predicting human behaviour in utility-based learning have shown impressive accuracy, even in complex visual tasks that require attention mechanisms [8]. However, these methods typically rely on the use of tabular RL and hand-crafted features that represent complex visual information using low-dimension one-hot encodings. Deep Reinforcement Learning (DRL) has demonstrated human-level performance on a range of visual based tasks, such as video games [9]. These neural network-based techniques reach super human level performance on a variety of tasks by leveraging a large amount of experience, often the equivalent of thousands of hours on a single task [9]. The model [1] presented in this work connects deep learning techniques in representation learning with predictions of utility that can be trained using RL to model human behaviour and visual representations in learning tasks.

2 Background

2.1 Variational Autoencoders

VAE models seek to learn representations of visual stimuli that preserve statistical structure. The closely related β-VAE method adds to the VAE a controllable information bottleneck regulated by an additional β parameter, to encourage generalization and robustness. As β increases the information capacity of the latent representation decreases, while when set to 0, information is unconstrained. β-VAE models have previously been used to model human choice in a visual categorization task [2] and visual decision making [6]. Results from this experiment provide evidence that systematic biases of human categorization can be explained by the information-constrained representations formed by β-VAE models.

Relatively less attention has been focused on the usefulness of these latent representations for utility learning tasks. The Uβ-VAE model presented in this work seeks to account for the interaction between utility-based learning and visual representation learning. The result is a deep learning based approach that is able to reflect the high generalization and sample efficiency of human learning, while also producing predictions of psychological representations that have applications beyond predicting behaviour, detailed in section 4.

2.2 Feature Reinforcement Learning

Feature Reinforcement Learning has been suggested as a plausible account for the high sample efficiency of human learning in visual learning tasks [8]. This method predicts the value of a stimulus as the sum of its constituent features; for example, a red square is valued as the sum of the features "red" and "square" individually. This encourages compositionality, reducing the amount of experience required to make predictions of utility with novel stimuli, improving generalization and sample efficiency, and better modelling human behaviour.

The FRL model was trained on a task in which participants were presented with three objects, each composed of three unique features: *shape* (square, triangle, circle), *color* (red, yellow, green), and *pattern* (dotted, wavy, hatched), and asked to predict the highest-utility stimulus, based on an observed reward. Critically, during blocks of trials ($\sim$20-25 stimuli), a single feature among the 9 possible features was indicative that a stimuli had a higher utility than the other two options. This single feature indicative of utility is the 'feature of interest', and the goal of a block of trials is to determine this feature and repeatedly select the option containing it. This allowed human participants to quickly generalize to stimuli pairs they had not seen before by leveraging the regularity of this utility function. Figure 1.A shows an example of the stimulus of red-wavy-square. A more complete description of the methodology is given in [8].

[1] Model code available at: github.com/TylerJamesMalloy/DisentangledVisualLearning

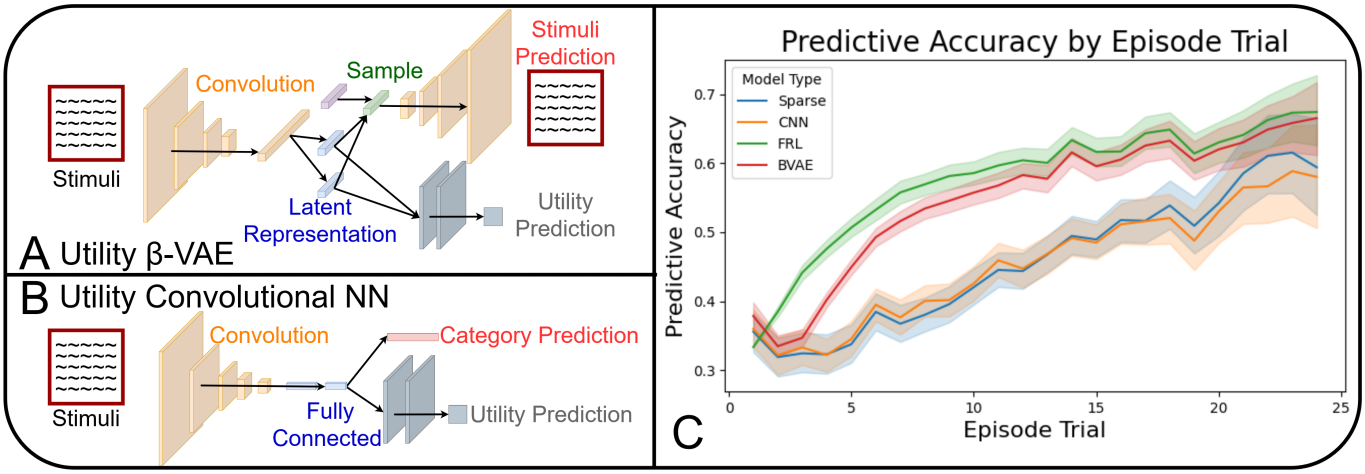


Figure 1: **A** Schematic of Utility β-VAE model with example input fed into convolutional layers to develop latent representations used both as input to utility prediction as well as sampled from to reconstruct original stimuli. **B** Schematic of CNN model used for comparison, structured similarly as the β-VAE based model, using the final fully connected layer to predict utilities. **C** Predictive accuracy of FRL, Utility β-VAE and 2 CNN models (sparse and standard).

3 Model

3.1 Utility β-VAE Model

To allow for utility-predictions based on visual stimuli, we developed a Utility β-VAE model (Uβ-VAE) that outputs both a stimulus reconstruction and utility prediction (see 1.A). This architecture is similar to previous methods that have been used in visual categorization tasks [2] and decision making [6] as well as deep RL methods that train based on abstract constrained representations of the state such as MuZero [9]. The Uβ-VAE model is trained using a loss function which balances the standard reconstruction error $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$ and model complexity $\beta D_{KL}\big(q_\phi(z|x)||p(z)\big)$ with the addition of utility error $\upsilon(\mathcal{U}(z) - u(x))$ through the following loss function:

$$\mathcal{L}(\theta, \mathcal{U}, \phi; x, z, \beta) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta D_{KL}\big(q_\phi(z|x)||p(z)\big) + \upsilon\big(\mathcal{U}(z) - u(x)\big)^2$$

In the model shown in 1.A, $\mathcal{U}$ represents the utility prediction module which outputs the prediction $\mathcal{U}(z)$ based on the latent representation z, and $u(x)$ represents the predicted utility from the supplied RL algorithm. This utility prediction error is weighted by the value υ, which allows for flexibility in the relevance of utility. The remaining terms are identical to those defined in the traditional β-VAE model. ϕ represents the parameters of the encoder $q_\phi(z|x)$, which defines the probability distribution over latent representations z given the stimulus x and θ represents the parameters of the decoder $p_\theta(x|z)$ of the stimulus x given the representation z. For a more complete description, see [4].

As mentioned, updating the predictions of utility relies on an error term based on the difference between the model predictions and a separate RL algorithm. In the results presented here these training utilities are supplied by the FRL algorithm. Training the Uβ-VAE model on utility predictions from the FRL algorithm allows for a comparison of our model against alternative deep learning approaches which are described in the following section.

3.2 Model Comparison

To provide a comparison with alternative DRL techniques, our model is compared against a similarly structured convolutional neural network (CNN) that is trained to predict the category of the stimulus (shape, color, and texture). This network (Figure 1.B) makes utility predictions based on the final layer of the network before the output categorization prediction. The network is trained in the same utility-weighted manner as the Utility β-VAE model. We also compare our model against a 'Sparse' CNN, which is trained using the same dimensionality for the utility prediction input as our model (6 values). Additionally, to emulate the properties of the β-VAE, we use Dropout, randomly zeroing some percentage (60%) of the NN nodes during training, which has been related to the information bottleneck principle [1].

In Figure 1.B models are compared in their accuracy of predicting human behaviour according to *predictive accuracy*, the probability that the model assigns to the action that was selected by the human participant. Note that the upper limit for all deep learning methods (BVAE, CNN, Sparse) is the accuracy of the FRL model that produces utility values used in training all DNN models. The Uβ-VAE model closely approaches the predictive accuracy of the baseline FRL, indicating that it has learned to predict the utility, but has significantly higher accuracy than the other deep learning methods.

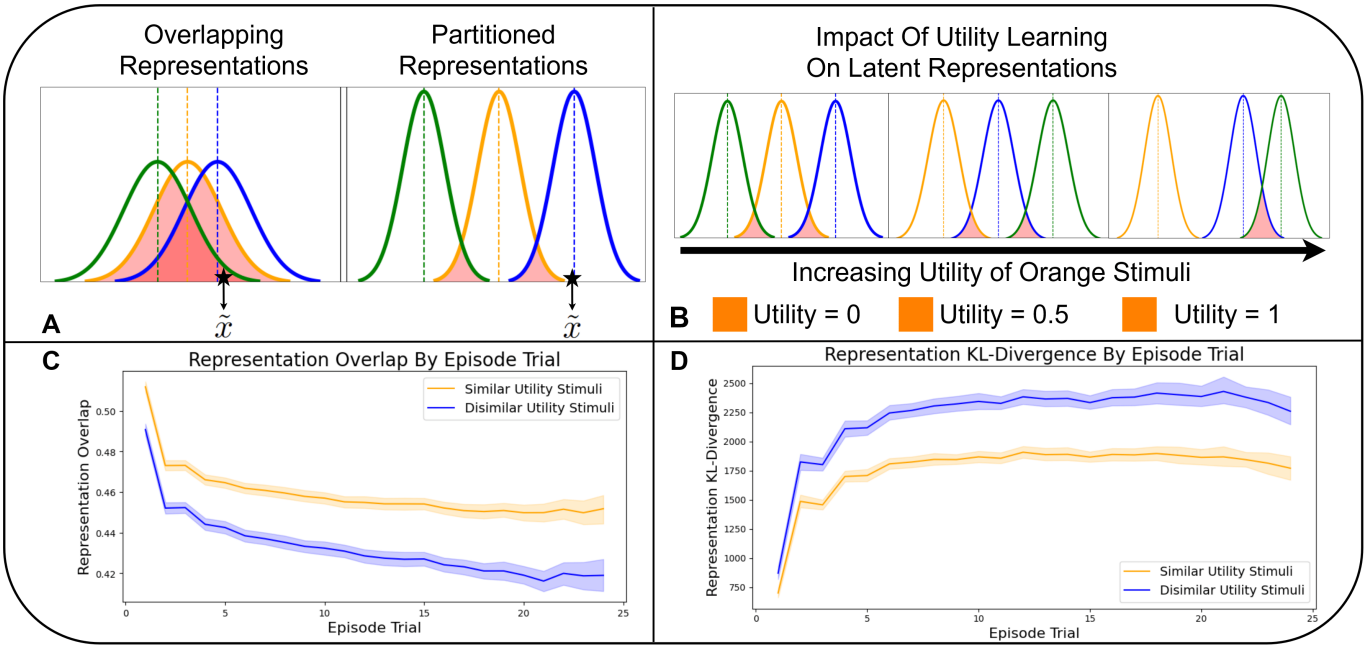


Figure 2: **A:** Demonstration of the relative likelihood that a data-point will be mislabeled (red area). Decreasing the likelihood of mislabelling results in a higher KL-divergence. The data-point $\tilde{x}$ drawn from the left overlapping blue distribution has a high likelihood of being mislabelled, this likelihood is decreased with the right partitioned distributions. **B:** Demonstration of the impact of learning utility values on latent representations. With all utilities as 0 (left) there is no preference for representation overlap. As the utility of the orange stimuli increases (middle) the model learns to prefer an overlap that reflects the acquired equivalence based on similar utility. Finally, as orange utility increases significantly (right), the model learns representations that only allow an overlap of stimuli with the same utility. **C:** Overlap of β-VAE latent representation distributions by episode trial, grouped by similar vs. dissimilar utility stimuli. **D:** Symmetric KL-divergence of β-VAE latent distribution overlap by episode trial, grouped by similar vs. dissimilar utility stimuli.

4 Modelling Results

The first method of analyzing the latent representations learned by Uβ-VAE uses information theoretic measures to explain the high generalization and robustness of the model. The β parameter in the β-VAE model limits informational complexity based on KL-Divergence. Overlapping latent representation distributions, such as those shown on the left side of Figure 2.A (based on a figure in [3]), have lower KL-Divergence scores and require less information to represent. However, a data point $\tilde{x}$ drawn from a distribution that overlaps with others is more likely to be incorrectly labelled as being drawn from a different distribution. This issue can be solved by partitioning representations, such as those shown on the right side of Figure 2.A, which increases the informational complexity of the learned representations. The loss function of the β-VAE model balances reconstruction accuracy and the complexity of learned representations. This has the effect of ensuring that neighboring stimuli in data space produce similar distributions in latent space [3].

In Uβ-VAE, representations are formed to balance reconstruction accuracy, informational complexity and utility-prediction error. This results in latent representations that depend on stimulus utility as well as visual structure. Previous research in acquired equivalence has demonstrated that stimuli representations can be updated based on categories within visual learning tasks [7]. The utility-based training of our model is motivated in part by a similar acquired equivalence, instead based on the utility that is associated with a visual stimuli in a learning task.

If this utility-based acquired equivalence exists in Uβ-VAE when predicting the behaviour of human subjects, it should be observable through a change in the similarity of latent representations. This expected change is demonstrated in Figure 2.B, which shows the hypothetical alteration of a learned latent representation as the utility associated with each stimuli is learned. As the utility associated with the orange stimuli increases, the cost associated with the orange stimuli being mistaken for either blue or green increases, and the overlap of these latent representation distributions changes. We test whether this effect takes place in Uβ-VAE by comparing the representation distribution overlap (Figure 2.C) and Kullback–Leibler divergence (Figure 2.D) between stimuli with similar utilities vs. stimuli with different utilities. The gap in overlap and distribution similarities based on utility provides evidence that the representations learned by our model have an acquired equivalence based on stimuli utility.

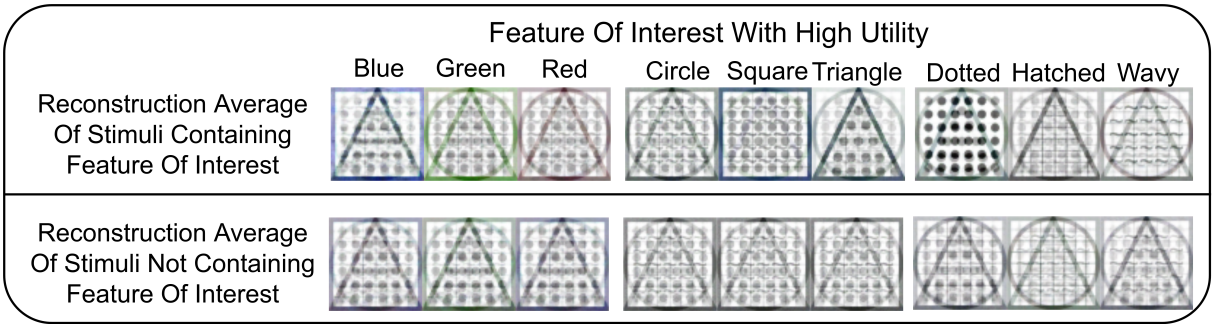


Figure 3: Average reconstructions from perturbed latent samples after training on each of the 9 features of interest, split into stimuli containing (top) and not containing (bottom) the given feature of interest. Note that the color features used (blue, green, red) differ from those used in [8] to maximize the reconstruction error between differently colored stimuli.

The final comparison of latent representations is done by visualizing the reconstructed images by first training our β-VAE model to predict a high utility for 1 of the 9 features of interest. After this, the learned latent representation for all stimuli is sampled from, with a small noise added. These noisy samples are used to reconstruct the stimuli, averaged over 1000 samples in the images shown in Figure 3. We compare the average reconstruction of stimuli that contain (top) and do not contain (bottom) the feature of interest. This visualization demonstrates that the learned representations of stimuli containing the feature of interest are more robust to noisy sampling. Specifically, when a particular color, shape, or texture is important for predicting utility, that feature tends to be better preserved in the stimulus reconstructions. This increased robustness is due to a partitioning of the representation space (as described in Figure 2.A) along the feature of interest, furthering the evidence of an acquired equivalence along the feature related to utility.

5 Conclusions

The Uβ-VAE model presented in this work quickly learns the stimuli utilities used by the FRL method to predict human performance in a complex visual task. This provides further support for the disentanglement objective as a plausible account of human visual representation formation, extending its application into visual utility learning. Compared with alternative deep learning methods, our approach better fits human behaviour while also producing latent representations that have been connected to biological visual representations in previous work.

Beyond predictive accuracy, the latent representations formed by our model can provide interesting insight into biological visual representations used in utility-based learning tasks. Analysis of both information theoretic and visualizations of these latent representations demonstrated evidence of a acquired equivalence of visually different stimuli based on their similar utility. This feature of Utility β-VAE representations opens up interesting possibilities for future research combining visual utility learning with additional tasks like change detection, categorization, and others.

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